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Health Technology Ecosystem Request for Information

DEPARTMENT OF HEALTH AND HUMAN SERVICES
CENTERS FOR MEDICARE & MEDICAID SERVICES

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Microsoft

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Company Information

Request	Information
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Introduction

NewWave Telecom & Technologies, Inc. (NewWave) is pleased to submit our response to this Request for Information. In the following sections, we demonstrate our extensive capabilities and relevant experience in delivering IT services of similar size, scope, and complexity.

NewWave is a Minority Owned Business with over 20 years of experience providing innovative IT, data management, and program support services to federal and healthcare clients. As a prime contractor on 95% of our projects, we have managed high-impact initiatives, including one of the largest healthcare data warehouses for the Department of Health and Human Services (HHS) Centers for Medicare & Medicaid Services (CMS). Our proven track record includes work with CMS and other federal agencies delivering secure, scalable solutions that drive mission success.

NewWave applies a disciplined, data-driven approach to ensure operational excellence, performance consistency, and continuous improvement across federal health programs. We implement an enterprise-wide Quality Management System (QMS) that integrates performance metrics and analytics to support informed decision-making and drive measurable gains in customer satisfaction. Through our leadership in launching Blue Button 2.0, NewWave pioneered the use of FHIR®-based APIs to expand beneficiary access to claims and coverage data, enabling secure, standards-based data exchange with over 75 third-party applications. As the original architect of Blue Button 2.0, NewWave is uniquely equipped to take this initiative to the next level - enhancing data availability, broadening functionality, and improving the overall user experience for Medicare beneficiaries. Similarly, our leadership on the Innovation, Development & Operations Services (IDOS) program at CMMI demonstrates our ability to deliver scalable, interoperable IT solutions, including the first FHIR-ready electronic Data Management Platform (eDMP) to support CMS's evolving digital ecosystem.

NewWave is well-positioned to support CMS in advancing its digital health ecosystem through our trusted partnership with Microsoft. As Microsoft's implementation partner for this effort, NewWave brings deep expertise in delivering secure, scalable, and interoperable health IT solutions that align with CMS's vision for patient-centered innovation, leveraging Microsoft's powerful cloud technologies—including Azure and its FHIR-enabled capabilities. NewWave will lead the implementation of solutions that enhance data accessibility, promote standards-based exchange, and empower beneficiaries and providers with actionable insights. Our shared commitment to FHIR as a foundational standard ensures alignment with CMS's interoperability goals and supports the seamless integration of clinical and administrative data across the care continuum.

Our team of certified professionals, including Microsoft-certified engineers and data specialists, excels in:

- **Data Management:** We design and manage complex data systems, ensuring seamless integration, security, and accessibility for large-scale healthcare operations.
- **Cloud Services:** As a Microsoft Gold Partner, we specialize in Microsoft Azure, delivering cloud migration, optimization, and management tailored to client needs.
- **Artificial Intelligence (AI):** Our AI and machine learning expertise, including custom NLP models, enhances analytics and fraud detection for smarter outcomes.
- **Software Development:** With CMMI Level 3 appraisals, we develop reliable, compliant applications.
- **DevSecOps:** We embed security and compliance into every phase of development, meeting stringent federal standards.

NewWave holds HITRUST e1 certification and leverages our experience to provide comprehensive IT support, ensuring continuity and risk mitigation. We offer complementary products like OnyxOS for healthcare interoperability and GeoAnalytica for geospatial insights, enhancing our service offerings. With a commitment to integrity and excellence, NewWave is poised to deliver trusted IT solutions that meet your mission-critical needs.

Team NewWave has responded to Sections B, C and E of this Request for Information.

A. Patients and Caregivers

This section is intended for all stakeholders to provide input on questions as they relate to use cases and workflows that involve patients and caregivers. While we certainly want patients and caregivers to answer questions in this section (and in other sections) from the patient/caregiver point of view, we also invite all stakeholders to provide their viewpoints on the patient/caregiver workflows.

1. Patient Needs

PC-1. What health management or care navigation apps would help you understand and manage your (or your loved ones) health needs, as well as the actions you should take?

a. What are the top things you would like to be able to do for your or your loved ones' health that can be enabled by digital health products?

b. If you had a personal assistant to support your health needs, what are the top things you would ask them to help with? In your response, please consider tasks that could be supported or facilitated by software solutions in the future.

PC-2. Do you have easy access to your own and all your loved ones' health information in one location (for example, in a single patient portal or another software system)?

a. If so, what are some examples of benefits it has provided?

b. If not, in what contexts or for what workflows would it be most valuable to use one portal or system to

access all such health information?

c. Were there particular data types, such as x-rays or specific test results, that were unavailable? What are the obstacles to accessing your own or your loved ones' complete health information electronically and using it for managing health conditions or finding the best care (for example, limitations in functionality, user friendliness, or access to basic technology infrastructure)?

PC-3. Are you aware of health management, care navigation, or personal health record apps that would be useful to Medicare beneficiaries and their caregivers?

PC-4. What features are missing from apps you use or that you are aware of today?

a. What apps should exist but do not yet? Why do you believe they do not exist yet?

b. What set of workflows do you believe CMS is uniquely positioned to offer?

PC-5. What can CMS and its partners do to encourage patient and caregiver interest in these digital health products?

a. What role, if any, should CMS have in reviewing or approving digital health products on the basis of their efficacy, quality or impact or both on health outcomes (not approving in the sense of a coverage determination)? What criteria should be used if there is a review process? What technology solutions, policy changes, or program design changes can increase patient and caregiver adoption of digital health products (for example, enhancements to data access, reimbursement adjustments, or new beneficiary communications)?

b. What changes would enable timely access to high quality CMS and provider generated data on patients?

PC-6. What features are most important to make digital health products accessible and easy to use for Medicare beneficiaries and caregivers, particularly those with limited prior experience using digital tools and services?

PC-7. If CMS were to collect real-world data on digital health products' impact on health outcomes and related costs once they are released into the market, what would be the best means of doing so?

2. Data Access and Integration

PC-8. In your experience, what health data is readily available and valuable to patients or their caregivers or both?

a. What data is valuable, but hard for patients and caregivers, or app developers and other technical vendors, to access for appropriate and valuable use (for example, claims data, clinical data, encounter notes, operative reports, appointment schedules, prices)?

b. What are specific sources, other than claims and clinical data, that would be of highest value, and why?

c. What specific opportunities and challenges exist to improve accessibility, interoperability and integration of clinical data from different sources to enable more meaningful clinical research and generation of actionable evidence?

PC-9. Given that the Blue Button 2.0 API only includes basic patient demographic, Medicare coverage, and claims data (Part A, B, D), what additional CMS data sources do developers view as most valuable for inclusion in the API to enable more useful digital products for patients and caretakers?

a. What difficulties are there in accessing or utilizing these data sources today?

b. What suggestions do you have to improve the Blue Button 2.0 API experience?

c. Is there non-CMS data that should be included in the API?

PC-10. How is the Trusted Exchange Framework and Common Agreement™ (TEFCA™) currently helping to advance patient access to health information in the real world?

a. Please provide specific examples.

b. What changes would you suggest?

c. What use cases could have a significant impact if implemented through TEFCA?

d. What standards are you aware of that are currently working well to advance access and existing exchange purposes?

e. What standards are you aware of that are not currently in wide use, but could improve data access and integration?

f. Are there redundant standards, protocols, or channels that should be consolidated?

g. Are there adequate alternatives outside of TEFCA for achieving widespread patient access to their health information?

PC-11. How are health information exchanges (HIEs) currently helping to advance patient access to health information in the real world?

a. How valuable, available, and accurate do you find the data they share to be?

b. What changes would you suggest?

c. Are there particular examples of high-performing HIE models that you believe should be propagated across markets?

d. What is the ongoing role of HIEs amidst other entities facilitating data exchange and broader frameworks for data exchange (for example, vendor health information networks, TEFCA, private exchange networks, etc.)?

PC-12. What are the most valuable operational health data use cases for patients and caregivers that, if addressed, would create more efficient care navigation or eliminate barriers to competition among providers or both?

a. Examples may include the following:

(1) Binding cost estimates for pre-defined periods.

(2) Viewing provider schedule availability.

(3) Using third-party apps for appointment management.

(4) Accessing patient-facing quality metrics.

(5) Finding the right provider for specific healthcare needs.

b. What use cases are possible today?

c. What should be possible in the near future?

d. What would be very valuable but may be very hard to achieve?

3. Information Blocking and Digital Identity

PC-13. How can CMS encourage patients and caregivers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would increasing reporting of complaints advance or negatively impact data exchange?

PC-14. Regarding digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 credentialing service providers (CSP)):

a. What are the challenges today in getting patients/caregivers to sign up and use digital identity credentials?

b. What could be the benefits to patients/caregivers if digital identity credentials were more widely used?

c. What are the potential downsides?

d. How would encouraging the use of CSPs improve access to health information?

e. What role should CMS/payers, providers, and app developers have in driving adoption?

f. How can CMS encourage patients to get digital identity credentials?

B. Providers

This section is intended for all stakeholders to provide input on questions as they relate to use cases and workflows that involve providers. While we certainly want providers to answer questions in this section (and in other sections) from the provider point of view, we also invite all stakeholders to provide their viewpoints on the provider workflows as appropriate.

1. Digital Health Apps

PR-1. What can CMS and its partners do to encourage providers, including those in rural areas, to leverage approved (see description in PC-5) digital health products for their patients?

To promote provider uptake—particularly in rural regions—CMS can:

- **Increase Funding and Policies to enhance Rural Broadband and Cloud Infrastructure:**

Dependable internet connectivity is essential for digital health. The presence and consistency of internet access continues to be a significant obstacle in rural healthcare. CMS can serve a catalytic function by enhancing support and funding for broadband infrastructure, especially via collaborations with telecom companies and federal broadband projects. Cloud-based systems are vital in rural regions with limited access to IT experts required for maintaining and supporting traditional on-premises systems. This connectivity enables more dependable, updated cloud-based EHRs that are safeguarded by the latest cybersecurity measures. Moreover, this secure framework will enable EHRs to interact, facilitating health record portability that enhances care and reduces the chance of overlooked information. Moreover, the wireless infrastructure will enable remote clinics and home care through wireless monitoring devices that enhance wellness and signal early indicators of illness, fostering proactive intervention to prevent expensive health crises. These tools will create the foundation for patients to retrieve their data and enhance transparency in their care. By collaborating, guaranteeing affordable and strong internet and cloud access "levels" the digital divide and enables modern digital health products to be viable in rural practice environments.

• **Link the use of authorized tools to financial incentives and value-driven care.** CMS needs to incorporate the use of digital health into the payment and quality incentive framework. For instance, CMS can directly incentivize providers for utilizing certified digital health solutions (virtual care, remote monitoring, AI-assisted decision tools, patient engagement applications, etc.) that enhance care and overall efficiency. Additionally, new billing codes and reimbursement structures can be established for the time invested in utilizing digital therapies or analyzing remote monitoring information. When enhanced patient outcomes from digital tools generate savings, providers ought to partake in those savings, thus making adoption financially appealing. Enhanced adoption speed might be realized through initial transformation funds or incentive payments for introducing remote monitoring in rural primary healthcare.

• **Create Collaborative Public–Private Partnerships for Expandable Solutions.** Collaboration between the public and private sectors can speed up innovation and lessen the pressure on individual providers. CMS can collaborate with tech firms and healthcare suppliers to offer secure, interoperable systems that rural providers may struggle to create or implement independently. An excellent instance is the Microsoft Cloud for Healthcare platform, which combines EHR data, remote monitoring, tools for care coordination, and analytics on a fully compliant cloud infrastructure. Collaborating with reliable industry partners allows CMS to implement pre-made solutions—developed on standards such as HL7® FHIR®—that integrate into current rural hospital processes. These partnerships would fulfill multiple needs simultaneously: improved care, reduced provider burnout, strengthened data interoperability, and enhanced cybersecurity. Such collaboration might establish a national framework for data exchange and remote patient monitoring with reduced barriers. CMS ought to function as a facilitator and stimulus, uniting top industry talent with rural providers to ensure that advanced digital solutions are implemented and backed by specialists, rather than allowing each clinic to independently develop solutions that cannot operate on the same interoperable framework.

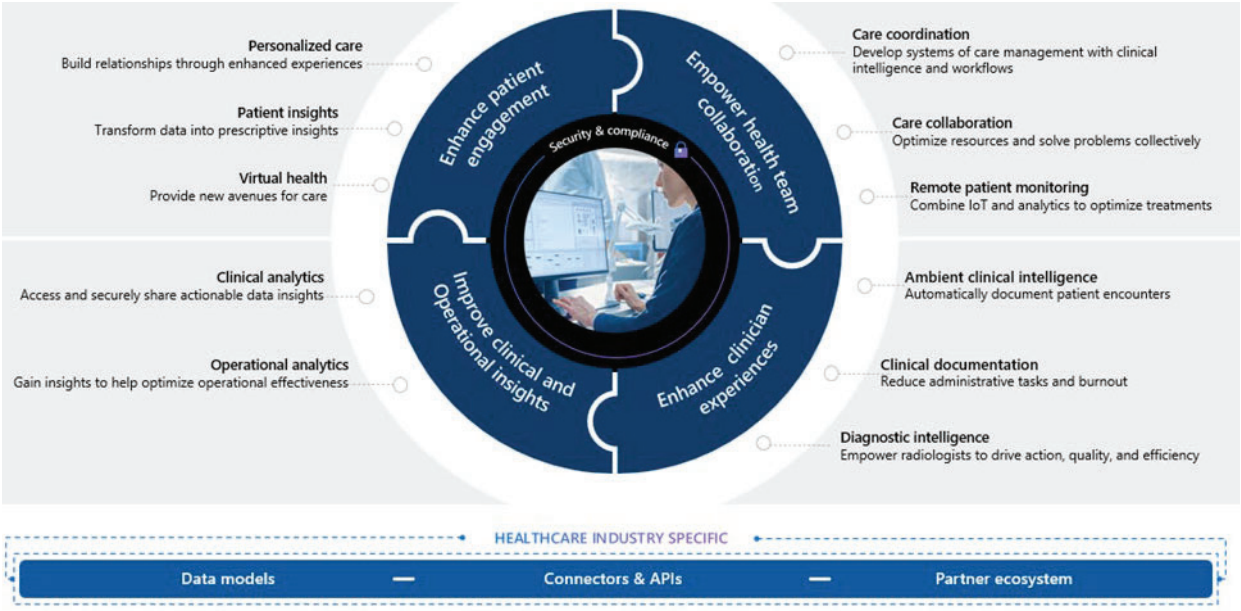

Figure: Microsoft Cloud for Healthcare

Microsoft's Cloud for Healthcare enhances patient accessibility, care coordination, and analytics, and is established on standards like FHIR and Azure Health Data Services. These collaborations can assist rural providers in evading the strain of constructing and upkeeping their own digital systems.

• **Offer Practical Training and Technical Support.** Despite having excellent tools at their disposal, providers require the expertise and assistance to utilize them efficiently. CMS can enhance and grow initiatives that assist staff in learning to implement digital health tools effectively. One strategy involves creating centers or networks—possibly in collaboration with academic medical centers or quality improvement organizations—to provide both on-site and virtual coaching for rural practices on various aspects, including establishing a patient portal and integrating new remote monitoring tools into their workflow. Virtual bootcamps, collaborative peer learning groups, and train-the-trainer initiatives are additional strategies. Collaboration between the public and private sectors should encompass technical support in areas such as workflow reengineering, data integration, and strategies for patient involvement. Simultaneously, CMS might create an AI-driven centralized helpdesk to aid in the configuration and resolution of issues related to digital products.

• **Obtain input and collaborative engagement from** providers - CMS must create ongoing feedback mechanisms with rural providers to grasp their specific challenges and work together on solutions. This may consist of advisory groups, test initiatives, questionnaires, and workshops focused on user-centered design. Additionally, engaging providers in collaboratively designing solutions boosts commitment. CMS might organize innovation workshops or hackathons with teams from rural providers to customize digital health workflows for their specific situation.

• **Concentrate on Addressing Providers' Most Urgent Issues through Digital Solutions.** CMS ought to stress how endorsed digital products directly address everyday challenges in clinical care, thus clarifying the value proposition. For instance, a significant challenge is the burden of administration and paperwork. AI-driven voice-recognition clinical documentation tools, like an AI scribe ("ambient clinical intelligence"), can reduce charting time; CMS ought to expedite its approval and contemplate subsidizing its expense. Another instance is care coordination for intricate patients: rural providers frequently send patients to far-off specialists and find it challenging to monitor the follow-ups. A secure messaging

platform or a shared care plan application that links local and specialist teams could significantly alleviate this issue – CMS can confirm they are reimbursable. Outcomes for patients and their satisfaction are also major concerns for providers. Facilitating the digital gathering of Patient Reported Outcomes (PRO) data (such as symptom assessments filled out by patients on their phones between appointments) and integrating that into providers' workflows can assist clinicians in rural clinics in tracking treatment effectiveness as it happens and responding sooner. CMS can facilitate this by making sure EHR certification criteria encompass PRO modules and by financing pilot initiatives that grant rural providers funding to integrate PRO dashboards for chronic disease management. In the end, CMS ought to promote digital solutions that show a decrease in hospital readmissions, enhanced access, and savings in clinician time.

a. What are the current obstacles?

• **Insufficient Technical Infrastructure in Rural Areas:** Rural areas frequently do not have the IT personnel and basic infrastructure—especially broadband internet and cloud services—that are crucial for implementing digital health solutions. Even with telehealth solutions in place, their effectiveness is limited by weak connectivity, old equipment, and insufficient IT support. This digital gap not only restricts patient access to online care but also hinders providers' capacity to utilize insights, incorporate the tools, and sustain the technologies.

• **Disconnected EHR Systems and Poor Interoperability:** A significant challenge continues to be the absence of interoperability among electronic health record (EHR) systems. Information saved in various formats—like scanned files, unstructured notes, and organized data fields—tends to be unreachable between systems. This fragmentation hinders the smooth sharing of patient information, diminishing the capability of digital health tools to facilitate coordinated, value-driven care.

• **Concerns from providers regarding liability, data privacy, and reimbursement:** Providers are cautious about the legal and financial implications tied to the adoption of digital health. Issues encompass ambiguous responsibility when suggesting digital tools, possible violations of patient confidentiality, and varying reimbursement policies. CMS could enhance its role in defining provider responsibilities and creating standardized reimbursement models to lessen ambiguity. These issues are particularly prominent in rural areas, where providers might not have legal or administrative assistance to manage these complexities.

• **Training Opportunities:** The absence of organized training initiatives for digital health tools and data literacy hinders the confidence of healthcare system providers and patients, consequently affecting adoption. Numerous rural providers function in environments with limited resources where time and financial support for professional growth are in short supply. The RFI indicates that CMS and its collaborators might consider investing in scalable, role-oriented training modules to enhance digital fluency and data literacy among healthcare providers and support personnel. Peer-to-peer learning communities and online training centers might also aid in closing this gap.

• **Overburdened Personnel:** Staffing deficiencies and fatigue are widespread in rural healthcare. Rural providers frequently find themselves overextended, managing various responsibilities with minimal assistance. Implementing new digital tools—even if they are advantageous—can seem more like an extra weight instead of a remedy. In the absence of focused implementation assistance or adjustments to workflows, digital health projects may be sidelined or scrapped.

• **Incorporating into workflows:** Even with digital tools accessible, incorporating them into current clinical workflows poses a significant challenge. Numerous digital health products lack a focus on usability for frontline workers. Providers require tools that match their current needs, workflows, documentation,

billing, and care coordination processes—not those that lead to redundant data entry or interrupt patient interactions. CMS might promote collaboration among developers and end-users in designing solutions and finance pilot initiatives that actively involve providers in the developmental phase.

b. What information should providers share with patients when using digital products in the provision of their care?

• **Objectives and advantages of the digital product:** Suppliers must clearly articulate the purpose of any digital health tool, detailing how it aids the patient's care pathway. A care navigation application could assist patients in finding specialists or booking appointments, whereas a health management application might monitor vital signs or medication compliance. These instruments are more than mere conveniences—they are intended to enhance results, decrease hospital admissions, and enable patients to engage more actively in their health. In countryside regions, where in-person healthcare may be scarce, digital solutions can fill essential voids by facilitating remote consultations, ongoing monitoring, and prompt interventions.

• **Safeguards for data privacy and security:** Patients should be made aware of how their information is safeguarded. This encompasses detailing encryption standards, protocols for identity verification, and adherence to HIPAA, FISMA, and various federal regulations. For example, CMS promotes utilizing FHIR-based APIs and identity verification partners to guarantee secure data sharing. Providers must also specify if the data is anonymized, pseudonymized, or fully identifiable, and the conditions under which it may be shared with external parties. Openness fosters trust—particularly in marginalized communities where doubts regarding digital monitoring might be greater.

• **The manner in which the product fits into their care plan:** Digital products must be viewed not as isolated tools but as integral parts of a patient's comprehensive care strategy. Providers must clarify how the product enhances current treatments, aids in care coordination, and facilitates immediate feedback exchanges between patients and clinicians. For instance, remote monitoring devices can transmit data straight to the EHR, enabling care teams to proactively modify medications or arrange follow-up appointments. This integration is particularly beneficial for overseeing chronic conditions such as diabetes or hypertension, where ongoing monitoring can avert complications.

• **Usage guidelines and support materials:** Simple, easy-to-understand instructions are crucial—particularly for patients with minimal digital skills. Providers ought to provide detailed guides, instructional videos, and support in multiple languages when feasible. Patients should be informed about how to access assistance, whether via a helpline, in-app chat, or community health workers. In rural regions, where broadband availability might be restricted, providers must guarantee that products operate offline or with minimal bandwidth, and that alternative support options (such as phone-based coaching) are accessible.

• **Advantages:** Service providers must clearly communicate the concrete advantages of utilizing the digital product. This could involve reduced in-person appointments, quicker access to treatment, enhanced medication compliance, and better health results. For instance, CMS has prolonged telehealth options until September 2025, enabling patients to obtain care from home without location limitations. Emphasizing these advantages can enhance patient involvement and acceptance, especially when presented in the context of convenience, safety, and empowerment.

c. What responsibilities do providers have when recommending use of a digital product by a patient?

• **Confirm the product's clinical suitability and approval:** Providers must take on the responsibility of ensuring that any digital health product they suggest is clinically safe and suitable for the patient’s unique

clinical situation, context, and care strategy. This involves coordinating the product's evidence foundation, regulatory standing (e.g., FDA approval), and clinical recommendations. For rural healthcare providers, where access to specialists can be restricted, this vetting procedure is even more essential.

• **Inform patients about correct usage and constraints**: After a digital product is chosen, providers need to make sure patients comprehend how to utilize it safely and efficiently. This involves providing training, explicit directions, clarifying capabilities and limitations of the tool, tackling digital literacy deficiencies, and making certain maintenance and support are straightforward. This must have clear step-by-step instructions (with images or video guides) and details on where patients can find support if they encounter problems – such as a customer service phone line, an in-app chat option, or help from community health workers. In regions with restricted internet access or for patients lacking digital literacy, it's crucial that tools operate offline or with low bandwidth, and provide alternative support (such as phone coaching). Training should also include information on data privacy, consent, and steps to take if the tool fails or produces unforeseen outcomes.

• **Track results and modify care plans as necessary:** Digital health tools need to function as part of a cohesive system. Providers should track patient outcomes and interactions with the tool, utilizing the generated data to guide and modify care plans. This encompasses monitoring compliance, alterations in symptoms, and any negative effects. In rural areas, where direct follow-up might be restricted, remote monitoring serves as an essential feedback mechanism.

• **Ensure adherence to HIPAA and CMS regulations:** Providers must confirm that all digital products utilized align with HIPAA and CMS data security and privacy standards. This entails confirming that the product has suitable protections for the storage and transmission of patient data, particularly in home or mobile environments. Providers need to stay informed about CMS's changing guidance regarding interoperability and data exchange, especially concerning rural health systems that could encounter infrastructure issues.

PR-2. What are obstacles that prevent development, deployment, or effective utilization of the most useful and innovative applications for physician workflows, such as quality measurement reporting, clinical documentation, and billing tasks? How could these obstacles be mitigated?

Obstacles consist of:

Fragmented Systems & Interoperability Gaps among EHR vendors: For instance, in the absence of standardized APIs among electronic health record (EHR) systems, a fragmented ecosystem will persist, making interoperability challenging and expensive. This also creates difficulties for third-party applications to integrate smoothly. This disintegration hampers innovation and delays the use of tools that could improve care. It also hinders providers from cross-covering at various locations using different EHRs. Might involve strategies for mitigation such as:

- **Implementation of FHIR-based standards:** Projects such as FHIR APIs and enforce data-sharing mandates.

- **Incentives driven by policy:** Regulatory agencies such as CMS can significantly influence by requiring or encouraging the adoption of open standards, data normalization, and interoperability frameworks.

- **Vendor collaboration:** Motivating EHR vendors to engage in collaborative industry groups can speed up the creation of shared APIs and lessen integration difficulties.

Substantial integration expenses and restricted IT assistance: Implementing new applications usually demands considerable initial investment in integration, testing, and continuous upkeep—resources that numerous healthcare organizations, particularly smaller practices, may be short on. Restricted IT resources exacerbate the issue, postponing execution and narrowing the range of innovations. Might encompass mitigation techniques such as:

- **Low-code/no-code platforms:** Solutions such as Microsoft Power Platform enable non-technical users to streamline workflows and connect systems without needing extensive technical knowledge.

- **Modular, cloud-centric solutions:** Embracing cloud-native apps with plug-and-play functionalities can lower infrastructure expenses and streamline deployment processes.

- **Shared services frameworks:** Regional health systems or coalitions can unite resources to assist one another by leveraging economies of scale and aiding smaller providers in implementing and sustaining advanced technologies.

Uncertainty in regulations regarding AI and automation: The changing regulatory environment concerning AI in healthcare leads to caution among both providers and vendors. Worries regarding compliance, liability, and the ethical application of AI can postpone or hinder valuable innovations. Might involve approaches for mitigation such as:

- **Clear governance structures:** Organizations are demanding robust, all-encompassing solutions that effortlessly integrate into clinical processes while maintaining ethical integrity. **Pilot programs and sandboxes:** Regulatory agencies can create supervised settings where innovative technologies can be evaluated securely, enabling stakeholders to collect data and enhance regulations.

- **Interdisciplinary cooperation:** Involving healthcare professionals, technology experts, ethicists, and regulatory authorities in the co-creation of AI solutions guarantees that tools are both functional and adhere to regulations.

Provider time, interest, perceived worth: Even the most sophisticated tools can falter if providers do not recognize their value or lack the time to implement them. Clinicians frequently feel burdened by administrative duties and may oppose new technologies that seem to increase complexity. Mitigation approaches might consist of:

- **Ambient and voice-activated documentation:** Tools such as Microsoft's Nuance DAX Copilot can minimize documentation time and mental strain, enabling doctors to concentrate more on patient care.

- **Provider-focused design:** Engaging practitioners in the creation and evaluation of innovative tools guarantees that solutions match actual workflows and challenges.

- **Showcasing ROI:** Presenting information on time efficiency, better documentation precision, and enhanced patient results can foster trust and encourage uptake.

Training and Change Management: Even with excellent technology, practices can struggle in execution because of insufficient training or support. Doctors and personnel frequently lack formal training on new

software and must independently learn how to navigate it. Recent feedback shows a shortage of established best practices, with providers seeking more direction from CMS on implementing digital tools. Mitigation: Connect the implementation of technology with training and assistance. Provide training sessions, user manuals, and collaborative learning groups to enhance confidence in utilizing new systems. Providing on-site and phone support while advocating for “super-users” aids teams.

PR-3. How important is it for healthcare delivery and interoperability in urban and rural areas that all data in an EHR system be accessible for exchange, regardless of storage format (for example, scanned documents, faxed records, lab results, free text notes, structured data fields)? Please address all of the following:

In urban and rural healthcare environments, the capability to securely share all types of data in an Electronic Health Record (EHR) system—regardless of being structured or unstructured—is essential not merely as a technical goal but as a fundamental necessity for compliant, equitable, efficient, and high-quality care provision. The availability of all data formats in EHR systems is not a privilege—it is essential for attaining genuine interoperability that results in improved outcomes. It guarantees that regardless of the location a patient obtains care—whether in a rural clinic or a city hospital—their health data is comprehensive, usable, and protected. This universality is crucial for enhancing results, minimizing disparities, and promoting the digital evolution of healthcare. It also facilitates the quick identification of emerging critical healthcare trends affecting millions, such as the early detection of drug side effects, interactions, and pandemics.

a. Current challenges in accessing different data formats.

• **Unconventional formats (e.g., scanned PDFs, faxes) are difficult to interpret:** Healthcare providers—particularly in rural locations—frequently depend on old communication techniques such as fax machines and scanned files. These formats are naturally unstructured, non-standard, and devoid of machine-readable metadata, creating challenges for electronic health record (EHR) systems to efficiently extract and integrate data. A scanned lab result can include vital patient data, but without optical character recognition (OCR) and natural language processing (NLP) or large language models (LLM) and agentic AI, that information stays trapped in an inaccessible image. This leads to missed chances, delays in coordinating care, heightened error risks, and restricts the capacity for analytics or managing population health assessments.

• **Legacy systems do not have modern APIs:** Numerous healthcare providers, especially in underfunded or smaller urban and rural settings, continue to use outdated EHR platforms that were not created with interoperability as a priority. These systems frequently do not have modern APIs, which are crucial for real-time data sharing between systems. In the absence of APIs, sharing data depends on manual exports, exclusive formats, or costly middleware options. This not only obstructs integration with modern systems but also restricts involvement in health information exchanges (HIEs) and national interoperability initiatives such as TEFCA (Trusted Exchange Framework and Common Agreement).

• **Obsolete and fragmented local systems:** Across urban and rural areas, numerous healthcare organizations still operate on-premises systems that are divided by department or function. These systems frequently fail to interact with one another, resulting in disjointed patient records and redundant efforts. For example, a rural clinic's billing software may not align with its clinical records system, necessitating manual data reconciliation by staff. This division decreases operational efficiency, heightens administrative workload, and may jeopardize patient safety because of partial or inconsistent data.

b. Impact on patient care quality.

Insufficient data results in overlooked diagnoses and unnecessary retests: In both city and countryside healthcare environments, inadequate interoperability and varying data formats hinder the continuity and quality of care. When healthcare providers do not have access to comprehensive, standardized patient data—such as lab results or medication histories—they are required to make decisions with restricted knowledge. This frequently results in diagnoses that are missed or postponed.

c. Technical barriers to full data accessibility.

- **Absence of OCR/NLP functionalities:** This hinders access to unstructured healthcare information. Numerous rural and urban practitioners continue to depend on scanned documents or handwritten notes. Without OCR, these records cannot be digitized by systems. Even when digitized, the lack of NLP hinders the extraction of essential insights from clinical notes, complicating integration into EHRs or HIEs and disrupting care coordination.

- **Irregular metadata labeling:** This hinders data interoperability. Systems frequently employ varying standards for labeling patient IDs, timestamps, or provider information. This lack of consistency results in discrepancies and loss of data during the transfer. Rural systems might experience challenges due to obsolete infrastructure and software, whereas urban systems encounter difficulties because of large data volumes—both situations impede smooth data sharing.

d. Cost or privacy implications of making all data formats interoperable.

- **Upfront costs for digitization and security:** Achieving interoperability in healthcare data demands a substantial upfront investment. Digitization entails transforming paper documents and standardizing formats across different systems, posing significant challenges for rural providers with limited resources. Security expenses encompass the deployment of cloud infrastructure, management of identities, and tools for compliance to adhere to regulations such as HIPAA and GDPR. Nonetheless, these investments are crucial to guarantee secure and effective data transfer, which will eventually reduce healthcare costs.

- **Increased risk of data breaches due to inadequate access controls:** Interoperability heightens the risk of overexposure if access controls are insufficient. Lacking advanced cybersecurity, stringent governance, and role-based permissions, sensitive health information may be vulnerable to unauthorized access. This poses especially high risks in rural regions with restricted access to technical resources and supervision. To guarantee privacy and security, it is essential to have strong modern cloud technology, protective measures, well-defined governance policies, and ongoing monitoring to avert misuse or unintentional exposure.

e. Priority level compared to other interoperability needs.

Interoperability is crucial as comprehensive data access is vital for fair and coordinated care. In urban and rural areas alike, easy access to various data formats allows providers to offer prompt, well-informed care, minimize redundancy, and address gaps. Lacking it, care becomes disjointed—particularly in underprivileged regions—compromising both quality and fairness. It also lays the groundwork for evidence-based prevention, enables individuals to reach their health objectives, and ensures transparency in options and competition.

PR-4. What changes or improvements to standards or policies might be needed for patients' third-party digital products to have access to administrative workflows, such as auto-populating intake forms, viewing provider information and schedules, and making and modifying an appointment?

CMS must:

- **Advocate for SMART on FHIR and OAuth 2.0 standards**: To facilitate secure, scalable, and interoperable access to administrative processes, CMS should officially endorse the implementation of SMART on FHIR and OAuth 2.0 as key standards. These technologies are already well-known for allowing third-party applications to securely access clinical data. Widening the application of administrative APIs—like those for scheduling appointments, managing provider directories, and handling intake forms—would establish a reliable and uniform foundation for both app developers and healthcare organizations. The final rule on CMS Interoperability and Patient Access promotes the utilization of FHIR-based APIs for the exchange of clinical data. Widening this to include administrative data would support CMS's overall objectives of decreasing burden and improving patient involvement. OAuth 2.0 guarantees that patients can permit applications to retrieve their data while maintaining security, an essential need for sensitive administrative tasks.

- **Mandate certified EHR vendors to provide administrative APIs:** CMS needs to require that certified EHR vendors release standardized administrative APIs as a component of their certification requirements. Although recent interoperability regulations have concentrated on clinical APIs, administrative processes are still disjointed and frequently unavailable to external applications.

By mandating vendors to adopt APIs for functions like appointment scheduling, provider availability, and filling out intake forms, CMS would stimulate a more transparent and competitive environment. This would enable patients to utilize digital tools they prefer for managing care logistics, enhancing convenience and lessening friction in care coordination.

This suggestion is reflected in the recent CMS Health Technology Ecosystem RFI, which requests feedback on how to address challenges in implementing innovative tools for physician workflows and patient involvement.

- **Promote sandbox environments for app testing:** To stimulate innovation and guarantee safety, CMS should advocate for or mandate the existence of sandbox environments where developers can assess third-party applications using administrative APIs. These settings ought to mimic actual workflows and feature fabricated data for intake forms, provider timetables, and appointment management systems. These sandboxes would reduce entry barriers for smaller developers, enhance testing for interoperability, and assist in uncovering usability or security problems prior to deployment. They would additionally aid in meeting CMS's objectives for lowering administrative burdens and enhancing digital health equity, particularly in underserved or rural regions.

Collectively, these policy changes would establish a stronger and more patient-focused digital health environment. They would also support CMS's continuous efforts to modernize healthcare provision through interoperability, as detailed in its recent regulatory and RFI initiatives.

2. Data Exchange

PR-5. Which of the following FHIR APIs and capabilities do you already support or utilize in your provider

organization's systems, directly or through an intermediary? For each, describe the transaction model, use case, whether you use individual queries or bulk transactions, and any constraints:

a. Patient Access API **via Azure API for FHIR.**

b. Standardized API for Patient and Population Services. **Used in Analytics and Care Coordination.**

c. Provider Directory API **integrated with Microsoft Entra (Azure Active Directory).**

d. Provider Access API. **Live/In Prod.**

e. Payer-to-Payer API. **Live/In Prod.**

f. Prior Authorization API. **Live/In Prod.**

g. Bulk FHIR—Do you support Group ID-based access filtering for population-specific queries? **Yes**

h. SMART on FHIR—Do you support both EHR-launched and standalone app access? What does the process for application deployment entail? **Both EHR-launched and standalone apps.**

i. CDS Hooks (for clinical decision support integrations). **Both EHR-launched and standalone apps.**

PR-6. Is TEFCA currently helping to advance provider access to health information?

a. Please provide specific examples.

b. What changes would you suggest?

c. What other options are available outside of TEFCA?

d. Are there redundant standards, protocols or channels or both that could be consolidated?

PR-7. What strategies can CMS implement to support providers in making high-quality, timely, and comprehensive healthcare data available for interoperability in the digital product ecosystem? How can the burden of increasing data availability and sharing be mitigated for providers? Are there ways that workflows or metrics that providers are already motivated to optimize for that could be reused for, or combined with, efforts needed to support interoperability?

PR-8. What are ways CMS or partners can help with simplifying clinical quality data responsibilities of providers?

a. What would be the benefits and downsides of using Bulk FHIR data exports from EHRs to CMS to simplify clinical quality data submissions? Can CMS reduce the burden on providers by performing quality metrics calculations leveraging Bulk FHIR data exports?

b. In what ways can the interoperability and quality reporting responsibilities of providers be consolidated so investments can be dually purposed?

c. Are there requirements CMS should consider for data registries to support digital quality measurement in a more efficient manner? Are there requirements CMS should consider for data registries that would support access to real-time quality data for healthcare providers to inform clinical care in addition to simplifying reporting processes?

3. Digital Identity

PR-9. How might CMS encourage providers to accept digital identity credentials (for example, CLEAR, ID.me, Login.gov) from patients and their partners instead of proprietary logins that need to be tracked for each provider relationship?

a. What would providers need help with to accelerate the transition to a single set of trusted digital identity credentials for the patient to keep track of, instead of one for each provider?

b. How might CMS balance patient privacy with convenience and access to digital health products and services that may lead to significant improvements in health?

PR-10. Regarding digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 CSPs):

a. What are the challenges and benefits for providers?

b. How would requiring their use improve access to health information?

c. What are the potential downsides?

d. What impact would mandatory credentials have on a nationwide provider directory?

e. How could digital identity implementation improve provider data flow?

f. Would combining FHIR addresses and identity improve data flow?

PR-11. How could members of trust communities [3] (for example, QHINs, participants and subparticipants in TEFCA, which requires Identity Assurance Level 2 (IAL2) via Credential Service Providers (CSPs)) better support the goals of reduced provider and patient burden while also enhancing identity management and security?

4. Information Blocking

PR-12. Should ASTP/ONC consider removing or revising any of the information blocking exceptions or conditions within the exceptions (45 CFR part 171, subparts B through D) to further the access, exchange, and use of electronic health information (EHI) and to promote market competition?

PR-13. For any category of healthcare provider (as defined in 42 U.S.C. 300jj(3)), without a current information blocking disincentive established by CMS, what would be the most effective disincentive for that category of provider?

PR-14. How can CMS encourage providers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would it advance or negatively impact data exchange?

C. Payers

PA-1. What policy or technical limitations do you see in TEFCA? What changes would you suggest to address those limitations? To what degree do you expect these limitations to hinder participation in TEFCA?

The Trusted Exchange Framework and Common Agreement (TEFCA) represents a hopeful advancement for universal health data exchange; however, we observe several constraints:

- **Insufficient Use of Modern APIs:** Existing TEFCA frameworks depend largely on document-based exchanges (e.g., IHE profiles) and have minimal incorporation of contemporary FHIR APIs. This may impede innovation and restrict direct access to data. We suggest incorporating support for FHIR-based exchange into TEFCA, which ONC has indicated is on the way. Microsoft's experience with FHIR (as the initial cloud provider offering a managed FHIR service) demonstrates that adopting FHIR RESTful APIs and bulk data can greatly enhance interoperability.

- **Identity and Trust Discrepancies:** TEFCA's method for managing participant identities is still developing. One limitation is the absence of uniform digital identity credentials among networks, making it challenging to establish trust and integrate new participants. We recommend utilizing high-assurance digital identities (e.g., credentials compliant with NIST 800-63 IAL2) for entities and individuals within TEFCA. Microsoft's Azure Active Directory (Entra ID) backs these assurance levels and could assist in establishing secure, federated identity for TEFCA participants, minimizing redundancy and manual identity verification efforts.

- **Incentives for Participation:** Payors might not recognize instant advantages compared to expenses. To enhance adoption, simplify onboarding and demonstrate value: for instance, allowing payers to utilize TEFCA as a unified entry point to access numerous providers and patients. Microsoft's technology helps lower technical obstacles – for instance, Azure Health Data Services can link to TEFCA networks via open APIs, enabling payers to integrate with TEFCA without needing to revamp their systems. Showcasing this seamless integration and instant access to extensive data will promote involvement.

- **Consequences for Participation:** Should these constraints go unresolved, payers may be reluctant to engage with TEFCA because of expenses or intricacies, obstructing network advantages. By updating TEFCA with FHIR APIs and strong identity trust, we anticipate greater engagement from payers. The HLS team at Microsoft and Health Research groups are engaged in interoperability standards and policy, such as taking part in HL7 FHIR Accelerators and providing guidance on data exchange frameworks, to ensure that initiatives like TEFCA are successful in application. By implementing the proposed modifications, TEFCA can become more accessible and user-friendly, reducing barriers to adoption.

PA-2. How can CMS encourage payers to accelerate the implementation and utilization of APIs for patients, providers, and other payers, similar to the Blue Button 2.0 and Data at the Point of Care APIs released by CMS?

CMS can significantly influence payers to embrace and implement APIs for sharing data with patients, providers, and other payers. Main recommendations consist of:

- **Emphasize & Reward Success Narratives:** Refer to CMS's own APIs (Blue Button 2.0 for patient Medicare information, and Data at the Point of Care (DPC) for healthcare providers) as examples. Demonstrate how these APIs enhance results – for instance, Blue Button 2.0 employs FHIR to provide beneficiaries with access to claims information, resulting in improved engagement. CMS might encourage payers (via regulations or innovation trials) to adopt comparable FHIR-based patient access APIs for their beneficiaries.

- **Utilize Standards and Offer Direction:** Numerous payers are already obligated to deploy the

Patient Access API and Provider Directory API according to CMS regulations. CMS must keep updating guidance and mandate compliance with HL7 FHIR implementation guidelines (CARIN Blue Button, Da Vinci PDex, etc.) to ensure payer APIs remain uniform throughout the industry. Microsoft provides resources to enhance compliance in this area – for instance, Azure Health Data Services (FHIR) can be readily set up to align with the CARIN Blue Button and Da Vinci PDex criteria for patient access and provider listings. This minimizes the development workload for payers.

- **Supply a Scalable Framework:** Certain payers might not have the means to create secure, scalable APIs. Through partnerships with cloud providers such as Microsoft, CMS could offer templates or environments for swift API implementation. Microsoft's technology suite features Azure API for FHIR, a managed service that addresses security, consent, and data format right from the start. This enables payers to share data with patients and providers rapidly, all while adhering to privacy regulations. Azure's FHIR service enables secure access through OAuth2/SMART on FHIR and was utilized by payers to comply with the 2021 CMS requirements.

- **Promote Payer-to-Payer Data Sharing:** In addition to APIs for patients and providers, CMS can advocate for APIs that facilitate data exchange between payers (ensuring patients' information transfers seamlessly when they change plans). Microsoft's FHIR tools facilitate the $member-match function for payer-to-payer data exchange, allowing for the automatic transfer of records during patient transitions between insurance providers. CMS may require or encourage the utilization of this capability to expedite payers' adoption of these interoperability features.

CMS can speed up industry progress by integrating policy tools (regulations, incentives) with assistance for contemporary API technology. Microsoft's cloud health offerings support this initiative by providing compliance-ready FHIR services and resources designed to meet CMS API specifications, reducing the obstacles for payers to adopt and effectively utilize these APIs for data sharing, rather than simply treating it as a checkbox task. Ultimately, simpler implementation and transparent ROI (enhanced member engagement, decreased administrative load) will accelerate API adoption throughout the payer community.

PA-3. How can CMS encourage payers to accept digital identity credentials (for example, CLEAR, ID.me, Login.gov) from patients and their partners instead of proprietary logins?

Payers can enhance user experience and security by accepting reliable digital identity credentials from patients and their authorized caregivers/partners, rather than requiring exclusive login accounts. Here's a method to accomplish this and the reasons it is beneficial:

- **Collaborate with Reliable Identity Providers:** Numerous patients currently possess secure digital identities (such as a Login.gov account for federal services or an ID.me identity for veterans or SSA services). Payers ought to implement federated login, enabling users to access the payer's portals or applications through their existing identities. From a technical standpoint, this is achievable through standard protocols (OAuth2/OpenID Connect). Microsoft's identity solutions simplify this integration: Azure AD External Identities (Azure AD B2C) allows the inclusion of Login.gov, ID.me, CLEAR, and various other identity providers for sign-in options. Indeed, Microsoft offers resources and tools for implementing ID.me integration in health contexts, demonstrating robust support for this method.

- **Enhanced Security and Compliance:** External identities such as Login.gov and ID.me meet NIST 800-63-3 IAL2/AAL2 standards, indicating they have completed thorough identity verification (e.g., validating government ID, facial recognition) and implement robust

authentication measures. By agreeing to these, payers instantly enhance the trust factor of user authentication without developing their own expensive solutions. Azure Active Directory aids in achieving compliance in this area – it is certified as FedRAMP High and adheres to NIST identity standards, enabling the enforcement of MFA and additional policies on federated logins when required.

- **Enhanced Patient Experience:** Implementing federated IDs allows patients to have one fewer username/password to remember. A Medicare beneficiary might utilize their Login.gov account for Medicare, Social Security, and also their supplemental insurer's portal – ensuring a seamless experience. Decreasing login obstacles can boost patient portal usage rates, facilitating improved access to health information. Research by Microsoft on user experience indicates that single sign-on and recognized credentials result in increased engagement frequency.

- **Implementation through Microsoft Technologies:** Microsoft's technology suite features Microsoft Entra Verified ID (for the distribution and acceptance of verifiable credentials), which can enhance the efficiency of identity sharing. For example, a trusted authority could issue a patient's verified identity as a secure credential that any payer can readily trust and accept. This approach, supported by standards such as DID (Decentralized Identifiers), is a key focus of ongoing Microsoft Health research aimed at enhancing identity interoperability. In the short term, leveraging Azure AD B2C federation features to support third-party logins offers a swift advantage for payers to eliminate proprietary logins.

In conclusion, Team NewWave suggests using a federated identity approach: payers utilizing government and third-party digital identities via open standards. This method enhances security confidence, alleviates pressure on patients, and is backed by Microsoft's cloud identity solutions currently (rendering it economical to apply). This approach allows payers to concentrate on health services instead of identity management, while patients benefit from a secure, convenient login experience across multiple systems.

PA-4. What would be the value to payers of a nationwide provider directory that included FHIR end points and used digital identity credentials?

A national provider directory featuring FHIR API endpoints and utilizing digital identity credentials would be extremely beneficial to payers. This directory would serve as a unified, reliable source of provider information (contact information, electronic service endpoints, and verified identities), resulting in multiple advantages:

- **Simplified Data Transmission:** Payers invest considerable resources in monitoring providers' locations, associations, and their FHIR API endpoints for data sharing. A consolidated directory accessible through FHIR API would enable payers to programmatically identify providers and their endpoints. This facilitates connections for claims, prior authorizations, or clinical data sharing. For instance, if a payer wants to access a provider's EHR FHIR server for patient information, the directory might offer the endpoint URL and the necessary credentials. Microsoft backs this vision by utilizing FHIR directory standards – for instance, the HL7 Da Vinci PDex Plan Network (PlanNet) Implementation Guide specifies how to depict networks and providers in FHIR. Azure Health Data Services can support a FHIR-based directory that meets these standards, allowing any payer system to integrate smoothly using FHIR queries.

- **Trust and Identity Verification:** Integrating digital identity credentials into the directory enhances reliability significantly. Every provider listing might be linked to a validated digital identity (for either the organization or the individual provider). For example, a provider may use a reliable credential

to validate their details, or the directory could show that a provider's identity was confirmed at a high assurance level. Microsoft's identity solutions (Azure AD and the forthcoming Verifiable Credentials service) can assist in this by allowing the distribution of digital certificates or credentials to providers. Payers would then be assured that a provider listed in the directory is legitimate, and could even utilize those credentials to create secure connections (e.g., mutual TLS or OAuth client credentials supported by a verified identity). This minimizes mistakes and deception in data exchange.

- **Benefit to Payers:** A national directory featuring verified identities would lower administrative costs – rather than each payer keeping its own provider database and continuously updating it, they can depend on the centralized directory. It would enhance interoperability – during the onboarding of new data-sharing scenarios, the initial step (identifying the correct endpoint and contact) is managed. It could additionally contribute to provider search tools for patients, establishing a consolidated source of truth. Microsoft's cloud platform is ideal for hosting and securing this vital national service, thanks to its scalability and adherence to security standards (HITRUST, HIPAA, etc.). Azure's strong role-based access management and monitoring allow for controlled updates to directory data, ensuring secure access for payers and other stakeholders.

- **Microsoft's Engagement and Abilities:** Microsoft has been participating in interoperability initiatives that support a provider directory, including the FAST (FHIR at Scale) project for a National Directory. Our technology suite offers all the components required: a FHIR database for the directory, API gateways, and identity management. For instance, Azure API Management might present the directory API with rate limiting and security features, whereas Azure AD authenticates the client identities that are accessing it. We also foresee integration with Microsoft Cloud for Healthcare tools – a payer utilizing our cloud could directly incorporate directory information into their workflows (claims processing, care management systems) via connectors or APIs, ensuring their provider data remains current.

In summary, a nationwide provider directory with standardized FHIR access and integrated digital identity would transform the landscape for payers. It is in accordance with Microsoft's dedication to open standards and trust in the healthcare sector. Team NewWave is prepared with the cloud infrastructure and knowledge to assist in creating and utilizing a directory that would greatly improve data exchange efficiency and minimize friction between payers and providers.

PA-5. What are ways payers can help with simplifying clinical quality data responsibilities of providers?

Payers can significantly lighten the clinical quality data load on providers by taking on greater responsibility for processing and reporting this data. An effective method is for payers to manage data aggregation and quality measure calculations through standardized formats such as FHIR. This enables providers to concentrate on delivering care, while payers handle the technical details of data submission and compliance.

A crucial approach is the implementation of digital quality measures (dQMs) based on interoperable standards like Clinical Quality Language (CQL). These facilitate automated, immediate reporting and decrease the inconsistencies and manual labor linked to conventional quality assessment. Through the adoption of common digital standards, payers and providers can enhance workflows and boost data precision.

Payers may also fund common infrastructure that enhances interoperability, like national provider directories, digital identity systems, and centralized data registries. These tools enhance access to quality data and lower administrative burdens for providers. Furthermore, incorporating remote patient

monitoring (RPM) data into quality reporting can improve care for high-risk groups and align with CMS objectives for patient empowerment.

Cooperation on data governance is just as essential. Payers and providers must collaboratively establish data quality standards and validation procedures to guarantee consistency and trust in the metrics reported. Ultimately, payers can complete the cycle by providing actionable insights—such as care gap notifications and performance dashboards—back to providers, integrating quality feedback right into clinical workflows to assist real-time decision-making.

a. How interested are payers and providers in EHR technology advances that enable bulk extraction of clinical quality data from EHRs to payers to allow them to do the calculations instead of the provider-side technology?

Interest in Bulk Data Extraction for Payers: There is an increasing focus on transferring the responsibilities of quality measurement from providers to payers, allowing for bulk clinical data extraction from Electronic Health Records (EHRs) to payers. We believe that both payers and providers are engaged, as long as the solution is secure, automated, and reliable. Enabling technology for this exists today:

- **FHIR Data Access in Bulk:** Current EHR systems and health clouds (like Microsoft's) adhere to the HL7 FHIR Bulk Data standard (Flat FHIR/NDJSON export) enabling the efficient export of extensive patient data collections. For instance, CMS's Data at the Point of Care API utilizes bulk FHIR to retrieve Medicare claims for a list of patients. Likewise, a payer might utilize bulk FHIR to request a provider's EHR for all information required for quality measures, instead of relying on providers to create reports. Microsoft Azure Health Data Services provides complete support for FHIR Bulk Data export and import, enabling the secure ingestion of comprehensive patient records from provider systems into a payer's analytics environment.

- **Scalable Analytics for Quality Metrics:** After acquiring the data, payers can utilize the Clinical Quality Language Engine and cloud-based analytics to calculate quality measures. Azure Synapse Analytics and Azure Machine Learning can leverage the consolidated FHIR data to compute quality metrics, such as HEDIS for STAR ratings more swiftly and reliably. Our platform is capable of managing large-scale real-world data and producing insights using AI/ML. This indicates that payers are capable of executing large-scale computations, freeing providers from handling these processes on their local systems. Providers are intrigued by this setup, particularly if it lessens the reporting load – Microsoft collaborates with partners on digital quality reporting trials where providers submit data through standardized FHIR APIs, and payers/CMS perform the calculations.

- **Quality and Standardization of Data:** To enable payers to calculate precise metrics, the data obtained from EHRs needs to be standardized. Microsoft plays a role here by providing our FHIR mapping and normalization tools. Azure Health Data Services can intake non-FHIR data (HL7 v2, C-CDA documents) from EHRs and transform it into FHIR resources through our interoperability pipelines. This guarantees that even if a provider's system isn't inherently FHIR, the payer continues to get uniform FHIR data. The outcome is reliable, high-quality data prepared for large-scale processing. Based on our experience, when this infrastructure is established, payers show considerable interest in adopting quality metrics (it offers them faster insights and greater control), and numerous providers are eager to transfer that responsibility.

b. In what ways can the interoperability and quality reporting responsibilities of providers to both CMS and other payers be consolidated so investments can be dually purposed? Are there technologies payers

might leverage that would support access to real time quality data for healthcare providers to inform clinical care in addition to simplifying reporting processes?

Streamlining Reporting Duties & Instant Data for Providers: Insurers can enhance the efficiency of quality and interoperability reporting by working with CMS and leveraging technology that fulfills two functions:

- **"Collect Once, Utilize Multiple Times" Principle:** We suggest that clinical data collected for a single payer's requirements should also meet CMS reporting needs, and the other way around. This signifies coordinating measure definitions and transport standards between CMS and additional payers. Microsoft backs standards such as FHIR-based quality measures (eCQMs) and Clinical Quality Language (CQL), enabling various parties to assess a single measure definition using the same data. Should CMS and payers embrace standardized digital measure definitions, a provider could submit data (or permit extraction) once, enabling multiple payers to compute their respective measures without further requests. Payers can utilize technologies such as Azure FHIR along with open-source measure calculation engines to execute any measure logic on their available data, including CMS measures, thereby ensuring that their investments in data pipelines are beneficial for all parties involved.

- **Immediate Quality Data Feedback:** Instead of relying solely on past reporting, payers can leverage technology to offer real-time or close-to-real-time quality information to providers, aiding in prompt clinical decision-making. For instance, as payers compile data, they may recognize care gaps or patients failing to meet specific quality standards and provide alerts or summaries to providers throughout the care process. This technology from Microsoft facilitates this two-way exchange. Through FHIR Subscriptions/Events, a payer's system can receive updates from a provider's EHR (provided patient consent and agreements are established). When, for example, a diabetic patient's lab results arrive, that information can flow to the payer (through Azure Event Hub integration). The analytics of the payer could refresh quality indicators instantly and provide the provider with an updated dashboard. Payers can utilize tools such as Power BI (which connects with Azure data stores) to develop quality performance dashboards that providers can access at any time. This offers instant benefits to providers – current performance metrics and patient-specific information – along with quarterly or annual reports for compliance.

- **Utilizing Current Networks and Minimizing Redundant Work:** Insurers could also work together through health information exchanges or consortiums to streamline provider reporting. For instance, if several payers utilize Azure as a reliable data exchange platform, a provider might upload data one time to a secure cloud site and allow multiple payers (and CMS) to access it. Microsoft's Azure confidential computing and data governance tools can guarantee that every party accesses only what they are allowed, safeguarding privacy and removing the need for individual uploads to each payer. This type of federated model, possibly aligned with TEFCA's network-of-networks, could significantly reduce unnecessary reporting pathways. Providers are liberated to concentrate on care, while payers (utilizing cloud automation) manage the data distribution to all authorized parties.

- **Technology for Instant Access:** To directly enable instant quality data access for providers, payers can implement FHIR APIs or applications for providers that highlight quality data. For example, a clinic provider might utilize a SMART on FHIR application (hosted and enabled by Microsoft Azure) that allows them to securely access the payer's most recent information about their patients. When a payer has reviewed claims and identified existing care gaps, the provider is able to access that information during the patient consultation. Microsoft is investigating these provider-centric tools within Microsoft's Cloud for Healthcare, acknowledging that completing the

feedback loop enhances outcomes, rather than merely meeting compliance metrics.

Through these methods, investments can have dual purposes: the infrastructure that payers create for their own requirements can also fulfill CMS's mandates and meet providers' information needs. Microsoft's technology suite (analytics, FHIR, cloud integration) is designed to facilitate this type of one-to-many data sharing and feedback mechanism securely. In essence, payers transform into collaborators in enhancing quality rather than merely accepting reports. This lessens the overall load on providers since the data is collected one time and shared by everyone, allowing providers to gain ongoing benefits from the data in patient treatment—transforming the process into one of collaborative care management rather than mere reporting.

PA-7. How can CMS encourage payers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would it advance or negatively impact data exchange?

CMS can motivate payers to file information blocking complaints through ONC's portal by emphasizing the significance and guaranteeing a secure, efficient process. Microsoft backs efforts to recognize and remove information blocking, as the unrestricted flow of data advantages all healthcare participants (patients, providers, and payers). These are our opinions:

- **Education and Guidance:** Initially, CMS needs to inform payers about what qualifies as information blocking in different situations (for example, a hospital that declines to provide discharge summaries to a payer's care management program unless a specific fee is paid could be subject to reporting). Numerous payers might be unaware that they can submit complaints or could be apprehensive about backlash from providers. Providing clear guidance and possibly anonymized complaint options may alleviate this. Microsoft can help by offering technology that monitors interoperability challenges – such as analytics that record instances when a payer's data requests to provider APIs repeatedly fail or are denied. These records (which can be obtained through Azure API Management or tailored solutions) can act as proof when filing a complaint, facilitating the process for payers to recognize and record blocking actions.

- **Efficient Submission Process:** Promoting reporting also involves simplifying it. If the Information Blocking Portal from CMS is user-friendly (possibly enabling bulk submissions or regular reporting), payers will be more inclined to participate. Microsoft's proficiency in user experience may enhance these processes – for example, allowing API-driven submission of complaints straight from a payer's IT system. If a payer's system (operating on Azure) identifies a trend of information blocking, it might automatically prepare a complaint report. This type of integration minimizes manual labor and promotes regular reporting as a component of operations.

- **Safeguarding Reporter Interests:** CMS might assure payers that disclosing information blocking will not adversely affect their relationships with providers over time. Highlighting that complaints can remain confidential and that enforcement is managed by regulators may make payers less reluctant. Additionally, CMS could take into account feedback mechanisms: if a payer submits a complaint resulting in action, it may be beneficial to notify that payer of the outcome. Understanding that reporting yields outcomes will encourage involvement.

- **Anticipated Outcome:** Should payers consistently lodge complaints, we expect an immediate rise in enforcement efforts, but a lasting decrease in information blocking behaviors. Illuminating problem spots will urge EHR vendors and providers to enhance compliance, leading to increased data exchange. This ultimately promotes data sharing by eliminating unnecessary obstacles – a beneficial result for the healthcare ecosystem. A possible issue is that a surge of complaints might

lead to a confrontational environment or burden regulators, which could be detrimental if not handled properly. Nonetheless, CMS can address this by promoting a collaborative approach: motivate payers and providers to handle issues directly when feasible and utilize the complaint process for ongoing problems. In other terms, present position reporting as a final backup to guarantee accountability, while simultaneously advocating for technological solutions that prevent blocking.

The optimal scenario is one in which data silos are dismantled intentionally. Our industry and policy teams (including the HLS group) have championed strong information blocking regulations along with a solid interoperability framework. Microsoft's cloud services facilitate easier data sharing and automate compliance—assuming all participants utilize interoperable systems (such as FHIR APIs, shared identity, etc., as mentioned earlier), true information blocking should be reduced. However, when it happens, a culture that encourages payers to report problems will hasten resolutions. Overall, motivating payers to report will enhance data sharing as it highlights challenges and promotes enhancements. Any possible adverse impacts (such as friction between payers and providers) can be addressed through transparent communication indicating that the objective is joint enhancement and patient advantage, rather than retribution.

We suggest that CMS explicitly urge payers to engage in the ONC complaint process for information blocking, offer resources and guarantees to facilitate this, and concurrently collaborate with the industry (including tech firms such as Microsoft) to address the underlying causes of blocking. This dual strategy will establish a positive feedback loop where increased reporting results in greater interoperability, which subsequently reduces the necessity for reporting as time progresses.

D. Technology Vendors, Data Providers, and Networks

This section is intended for all stakeholders to provide input on questions as they relate to use cases and workflows that involve technology vendors, data providers, and networks. While we certainly want technology vendors, data providers, and networks to answer questions in this section (and in other sections) from their point of view, we also invite all stakeholders to provide their viewpoints on the technology vendor, data provider, and network use cases as appropriate.

1. Ecosystem

TD-1. What short term (in the next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?

TD-2. Regarding CMS Data, to stimulate developer interest—

a. What additional data would be most valuable if made available through CMS APIs?

b. What data sources are most valuable alongside the data available through the Blue Button 2.0 API?

c. What obstacles prevent accessing these data sources today?

d. What other APIs should CMS and ASTP/ONC consider including in program policies to unleash innovation and support patients and providers?

2. Digital Identity

TD-3. Regarding digital identity implementation:

a. What are the challenges and benefits?

b. How would requiring digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 CSPs) impact cybersecurity and data exchange?

c. What impact would mandatory use of the OpenID Connect identity protocol have?

3. Technical Standards and Certification

TD-4. How can CMS better encourage use of open, standards-based, publicly available APIs over proprietary APIs?

TD-5. How could a nationwide provider directory of FHIR endpoints improve access to health information for patients, providers, and payers? Who should publish such a directory, and should users bear a cost?

TD-6. What unique interoperability functions does TEFCA perform?

a. What existing alternatives should be considered?

b. Are there redundant standards, protocols or channels or both that should be consolidated?

TD-7. To what degree has USCDI improved interoperability and exchange and what are its limitations?

a. Does it contain the full extent of data elements you need?

b. If not, is it because of limitations in the definition of the USCDI format or the way it is utilized?

c. If so, would adding more data elements to USCDI add value or create scoping challenges? How could such challenges be addressed?

d. Given improvements in language models, would you prefer a non-proprietary but less structured format that might improve data coverage even if it requires more processing by the receiver?

TD-8. What are the most effective certification criteria and standards under the ONC Health IT Certification Program?

TD-9. Regarding certification of health IT:

a. What are the benefits of redefining certification to prioritize API-enabled capabilities over software functionality?

b. What would be the drawbacks?

c. How could ASTP/ONC revise health IT certification criteria to require APIs to consistently support exchanging data from all aspects of the patient's chart (for example, faxed records, free text, discrete data)?

d. What policy changes could CMS make so providers are motivated to respond to API-based data requests with best possible coverage and quality of data?

e. How could EHRs capable of bulk data transfer be used to reduce the burden on providers for reporting quality performance data to CMS? What capabilities are needed to show benefit? What concerns are

there with this approach?

TD-10. For EHR and other developers subject to the ONC Health IT Certification Program, what further steps should ASTP/ONC consider to implement the 21st Century Cures Act's API condition of certification (42 U.S.C. 300jj-11(c)(5)(D)(iv)) that requires a developer's APIs to allow health information to be accessed, exchanged, and used without special effort, including providing access to all data elements of a patient's electronic health record to the extent permissible under applicable privacy laws?

TD-11. As of January 1, 2024, many health IT developers with products certified through the ONC Health IT Certification Program are required to include the capability to perform an electronic health information export or "EHI export" for a single patient as well as for patient populations (45 CFR 170.315(b)(10)). Such health IT developers are also required to publicly describe the format of the EHI export. Notably, how EHI export was accomplished was left entirely to the health IT developer. Now that this capability has been in production for over a year, CMS and ASTP/ONC seek input on the following:

a. Should this capability be revised to specify standardized API requirements for EHI export?

b. Are there specific workflow aspects that could be improved?

c. Should CMS consider policy changes to support this capability's use?

4. Data Exchange

TD-12. Should CMS endorse non-CMS data sources and networks, and if so, what criteria or metrics should CMS consider?

TD-13. What new opportunities and advancements could emerge with APIs providing access to the entirety of a patient's electronic health information (EHI)?

a. What are the primary obstacles to this?

b. What are the primary tradeoffs between USCDI and full EHI, especially given more flexible data processing capabilities today?

TD-14. Regarding networks' use of FHIR APIs:

a. How many endpoints is your network connected to for patient data sharing? What types, categories, geographies of endpoints do you cover? Are they searchable by National Provider Identifier (NPI) or organizational ID?

b. How are these connections established (for example, FHIR (g)(10) endpoints, TEFCA/Integrating the Health Enterprise (IHE) XCA, or proprietary APIs)?

c. Do you interconnect with other networks? Under what frameworks (for example, TEFCA, private agreements)?

TD-15. Regarding bulk FHIR APIs:

a. How would increased use of bulk FHIR improve use cases and data flow?

b. What are the potential disadvantages of their use?

TD-16. What are the tradeoffs of maintaining point-to-point models vs. shared network infrastructure?

a. Do current rules encourage scalable network participation?

b. What changes would improve alignment (for example, API unification, reciprocal access)?

TD-17. Given operational costs, what role should CMS or ASTP/ONC or both have in ensuring viability of healthcare data sharing networks, including enough supply and demand, that results in usage and outcomes?

5. Compliance

TD-18. Information blocking:

a. Could you, as a technology vendor, provide examples for the types of practices you have experienced that may constitute information blocking. Please include both situations of non-responsiveness as well as situations that may cause a failure or unusable response?

b. What additional policies could ASTP/ONC and CMS implement to further discourage healthcare providers from engaging in information blocking practices?

c. Are there specific categories of healthcare actors covered under the definition of information blocking in section 3022(a)(1) of the Public Health Service Act (PHSA) that lack information blocking disincentives?

TD-19. Regarding price transparency implementation:

a. What are current shortcomings in content, format, delivery, and timeliness?

b. Which workflows would benefit most from functional price transparency?

c. What improvements would be most valuable for patients, providers, or payers, including CMS?

d. What would further motivate solution development?

E. Value-Based Care Organizations

This section is intended for all stakeholders to provide input on questions as they relate to use cases and workflows that involve value-based care organizations. While we certainly want value-based care organizations to answer questions in this section (and in other sections) from the value-based care provider point of view, we also invite all stakeholders to provide their viewpoints on the value-based care workflows as appropriate.

1. Digital Health Adoption

VB-1. What incentives could encourage APMs such as accountable care organizations (ACOs) or participants in Medicare Shared Savings Program (MSSP) to leverage digital health management and care navigation products more often and more effectively with their patients? What are the current obstacles preventing broader digital product adoption for patients in ACOs?

CMS faces a challenge with data model attribution. Data model attribution is commonly utilized in the

American healthcare system as a successful method for enhancing health results in the population. The Society of Actuaries describes attribution as the method of linking patients to particular provider entities that oversee most of the patient's care. The provider organization is responsible for both the expenses of healthcare and the quality of care provided to patients. In a healthcare system based on attribution, accountability and efficiency are essential as the expenses of a patient's treatment, care history, and health outcomes are monitored by a specific healthcare provider. Data models analyze a patient's electronic health records, claims information, and demographics to identify the suitable clinic or entity responsible for their healthcare. Consequently, data model attribution corresponds with value-based care, which promotes quality and cost-effectiveness for reimbursements rather than the quantity of services provided to patients. Attribution of data models in healthcare represents a transition from the conventional fee-for-service system in Medicare to a value-driven approach to care. Value-based care is a framework that focuses on generating value for patients, accomplished only when an individual's health enhances. In that context, compensation for provider entities is associated with patient outcomes rather than the quantity of services rendered. The focus is on the results, such as a decrease in the number of clinic appointments and an enhancement in the control of chronic illnesses like diabetes. This operational model demands precise assignment of patients to healthcare providers capable of delivering high-quality care for effective cost control and reimbursement accountability.

Organizations recognized as Accountable Care Organizations (ACOs) or suppliers of bundled payment programs depend on patient attribution to assess their financial obligations. Comprehending the individual who assumes risk in a value-based contract is essential for successful execution. Data model attribution assists in pinpointing the risk pool for every provider, outlining their obligation to guarantee quality care for the patients in their assigned list according to the agreements. Furthermore, accurately attributing patients ensures that providers possess the required patient information to allocate sufficient resources to improve health outcomes. Involved providers can monitor patients' health patterns, identify high-risk individuals, and execute successful care plans.

Shifting to value-based care requires proper patient assignment to aid in providing high-quality and economical care. The creation and execution of effective attribution models often pose challenges because of problems such as inconsistent data and the absence of standardized frameworks. The rollout of value-based care, incorporating patient attribution, has been gradual due to shifts in policy and regulations, insufficient data standards, challenges with data speed, conflicting incentives, and a lack of trust between stakeholders. Due to these obstacles, the precision of patient attribution is adversely impacted, affecting both reimbursements and patient results.

Scheduling of Patient Attribution Models

As previously mentioned, patient attribution refers to the allocation of patients to a particular healthcare provider, who then assumes responsibility for their care. The assignment of patients to particular provider entities arises from the demand for responsibility regarding the quality and cost of care in value-based healthcare payment models. Over 170 attribution models exist, with at least 30 actively in use. The large variety of attribution models being created stems from the difficulty of establishing a singular, effective system for calculating payments, monitoring care coordination, and reporting patient results. Every category of model yields a distinct output depending on the sources of data utilized. Nevertheless, the primary goal across all models is to ensure precise attribution to improve population health management and mitigate risk for providers. Without an appropriate attribution model, healthcare providers face difficulties in recognizing their patients or assigning adequate resources to address their conditions, resulting in inefficiencies within the healthcare system.

Patient Assignment Methodology

Different factors are employed to assign patients to a particular provider, each having its own benefits and drawbacks. The three typical attribution methods include geographical, plurality of care, and patient self-reported attribution. With geographical attribution, patients are allocated to particular facilities depending on how close they live. Information like zip code is utilized to match patients with a nearby provider. A significant drawback of this method is that it does not consider the number of times each patient visits a particular healthcare facility. The visit-based attribution faces no such difficulty as it employs algorithms to allocate patients to providers using claims data. The provider that filed the most claims for a specific patient during a designated timeframe is designated as that patient's primary care provider, which is a standard practice for many government-funded healthcare programs. Although visit-based attribution relies on data and is simple, its success in correctly linking patients to providers greatly relies on the quality of data present at the provider level. A frequent assumption in the model is that the provider with the highest number of claims is the primary provider, which might not be correct. For example, a specialist treating a chronic illness might file more claims than the patient's primary care doctor, resulting in misattribution. Additionally, the retrospective approach is utilized in the majority of attribution methods, resulting in a delay in acknowledging provider-patient connections.

The third attribution method is self-reporting by patients, which is optimal. In this method, the patient directly chooses or verifies their main healthcare provider. Although this approach effectively allocates patients to providers, it depends on the patient's willingness to self-report and their comprehension of the different providers. A patient might not recognize the best provider for their primary health needs, resulting in a disconnect with the objectives of value-based care. Furthermore, personal choices can fluctuate frequently because of individual tastes or relocation, requiring ongoing updates. Consequently, each of the three patient attribution methods has constraints that could impede their effectiveness in creating an appropriate model for linking patients to providers.

Attribution of Data Models in Alternative Payment Frameworks

Attribution is used in two main alternative payment models to link or assign patients to providers: the episode-based and patient-based payment models. In the episode-based payment approach, one payment is utilized to cover multiple predefined services delivered over a specific timeframe (Society of Actuaries, 2018). This method is typical in maternal care, which is categorized as a set of specified services. The provider is accountable for delivering all services in every episode within a designated timeframe. In contrast, the patient-centered payment model focuses on the services provided to an individual patient. Compensation is given to a provider for all services rendered within a specified timeframe. ACOs generally adopt this method, receiving one payment to manage care delivery for every patient. The data-model attribution is important in both payment models as it establishes the foundation for allocating profits or charges to the provider organizations. Standardized metrics serve to allocate responsibility to the care providers and then create the formula for determining gains and fines to be distributed or imposed, respectively. Providers typically concentrate on the data model employed in these processes to guide their choices regarding enrollment in an attribution program.

Effect on Value-Centered Care

As mentioned previously, patient attribution plays a crucial role in value-based care to impact both financial and clinical responsibility. Cantor (2020) asserts that accurate attribution is essential in any initiative aiming to implement value-based care or assess quality. By utilizing data models to allocate patients, providers can pinpoint the group that constitutes their risk pool. Value in healthcare relies on two main elements: the health outcomes of the patient and the expenses incurred to achieve those outcomes. In every case, appropriate attribution is essential, establishing a need for efficient use of data

models to guarantee that financial rewards accurately align with performance indicators. Providers offering quality, affordable care gain from shared savings initiatives and incentive rewards. Those who do not comply with care quality standards may encounter financial repercussions. Data model attribution enables healthcare organizations such as Medicare to ensure providers are responsible for their performance and patient results. Attribution assists service providers such as ACOs in clarifying their duty regarding the care management strategies they employ. The organizations utilize attribution lists to comprehend their patients' requirements, facilitating the efficient coordination of care to enhance value for patients. Consequently, data model attribution promotes value-based care by syncing financial rewards with the quality of care provided.

Effect on Population Health Management

Population health management aims to enhance health results within a defined population. To manage population health effectively, accurate and dependable attribution of patients to healthcare providers is essential. They emphasize that the aim of population health is to improve preventative care to lower the financial and health impacts of different infections. Data model attribution aids in dividing the population into categories considering factors such as risk elements and health issues. Precise attribution thus aids preventive care approaches, a fundamental necessity in managing population health. Significantly, a patient diagnosed with diabetes can benefit from proactive measures like routine screenings, medication oversight, and lifestyle advice, enhancing their health results in the long run. Furthermore, precise attribution aids in resource distribution by allowing the identification of the patient group to guide funding needs, care teams, and priority sectors for disease prevention and management. The implementation of appropriate preventive care for the identified patient group lowers the rate of hospitalizations and healthcare expenses. ACOs gain from enhancements in population health management made possible by data model attribution, allowing them to monitor the impact of the interventions they implement. Providers can evaluate if particular strategies result in better patient outcomes, enabling them to adjust their methods and embrace the most effective practices.

Effect on Healthcare Quality Documentation and Performance Assessment

Value-based care depends on key performance indicators (KPIs) to evaluate the quality of care delivered by different healthcare providers. Attribution of data models offers an appropriate structure for assessing the quality of care given to particular patients according to the results. The quality of reports produced by healthcare systems influences the capacity to assess healthcare effectiveness, a problem addressed by the attribution process. Government initiatives such as the Medicare Shared Savings Program (MSSP) and the Merit-Based Incentive Payment System (MIPS) depend on patient attribution to assess provider performance ratings. In these situations, misattribution of patients might result in improper distribution of financial rewards or sanctions. Furthermore, precise attribution reinforces the credibility of public health reporting. Healthcare organizations utilize performance metrics to recognize trends, evaluate issues in care provision, and create specific interventions. Misassigned patients can skew these insights, hindering the implementation of successful quality improvement efforts.

Medicare's Issues with Patient Assignment

Although patient attribution is crucial in value-based care, its execution has been difficult within the healthcare system. Attributing patients is becoming more crucial for evaluating care quality and compensating providers. Nonetheless, the procedure continues to be difficult. Cantor (2020) contends that key obstacles to patient attribution arise from various issues, such as the absence of a specific primary care physician, patients obtaining care from several physicians across different networks, and inconsistencies in data quality and sources regarding patient visits to healthcare facilities. All of these problems obstruct the precise allocation of patients to providers, resulting in incorrect payment of financial

incentives or imposition of penalties and hindering the evaluation of the quality of care given to patients.

Concerns about Data Quality and Consistency

The quality of data is fundamental to every patient attribution model. Numerous inaccuracies exist in healthcare data that affect the secondary use of that data, which in this instance would involve patient attribution. Data problems adversely impact the standard of care given to patients. The quality of data restricts the complete use of healthcare information in addressing issues within the industry, such as managing healthcare expenses. Issues with data quality arise from utilizing various benchmarks to assess the quality of data within the health information system. These findings in literature indicate that data quality is a significant concern in the healthcare industry. In patient attribution, necessary data is frequently gathered from various sources, including electronic health records (EHRs), insurance claim data, and patient self-reported information through digital or manual forms. Each of these sources might employ various techniques to gather, retain, and present information. An important illustration is the application of claims data for patient attribution. Although this data offers insights into a patient's use of healthcare services, it may not accurately reflect the patient-provider relationships for attribution reasons. Delays in claims data related to billing cycles hinder the ability to identify which provider is actively engaged in delivering primary care to a patient.

In addition to claims data, electronic health records lack standardization among providers and healthcare systems, making accurate attribution difficult when patients receive care from various providers. Variations in patient documentation hinder the attribution process, leading to inconsistencies. Furthermore, data reported by patients themselves, commonly seen as a benchmark in patient attribution, is not consistently reliable because of personal biases and the utilization of various services from different providers. Consequently, the absence of standardized data collection techniques in healthcare results in attribution models not producing valuable insights for forming patient-provider relationships, jeopardizing accurate attribution.

Frequent Changes in Providers

Patients frequently transition between various care providers in the healthcare industry. Cantor (2020) states that seeing various doctors across different networks complicates attribution since accurately identifying consistent patient-provider relationships is challenging. Most attribution models depend on the sustained continuity of care to outline these connections. Nevertheless, patients frequently have to switch healthcare providers because of reasons such as changes in insurance plans and moving. For numerous individuals, their insurance coverage is supplied by their employers, which can vary during yearly renewals. During renewals, networks for insurance coverage may be altered, requiring patients to transition to a doctor within the new coverage network. As a result, an interruption in the continuity of care takes place, obstructing effective attribution in the majority of models. A comparable issue emerges from patient transfers, as healthcare providers must revise attribution models to reflect the shifts to different geographical locations. Seasonal employees and military households might not have a stable care location, resulting in regular changes in providers. This issue frequently impacts geographical-attribution models, necessitating constant updates. Consequently, frequent changes in providers, whether caused by shifts in insurance plans or patient relocations, hinder long-term continuity of care, which is crucial for precise patient attribution.

Methodological Constraints

The approaches employed in assigning patients influence how they are matched with providers and the resulting health outcomes. Both retrospective and prospective approaches have their own limitations. The retrospective method, commonly employed in attribution models, relies on past data regarding

healthcare use, which might not represent the existing patient-provider interaction. Utilizing historical data increases clarity but restricts adaptability for altering patients. Significantly, a patient might transition to specialist care this year while having utilized a primary care physician's services the previous year, resulting in a flawed performance assessment as the change remains unrecorded during attribution. The provider accountable for present health results might not obtain the financial rewards because of misattribution. Prospective models lack this challenge because they facilitate proactive planning; however, they may be inaccurate if patients change providers unexpectedly. The model lacks a method to consider real-time patient movement, potentially resulting in lapses in healthcare coordination. Tackling these limitations in every model is essential to improve the efficacy of patient attribution.

Executing Efficient Patient Attribution

As mentioned in the earlier section, patient attribution poses challenges stemming from various factors that impact the identification of patient-provider connections. Tackling these challenges is crucial for the healthcare system to attain the anticipated advantages of patient attribution. Here are several suggested modifications.

Standardizing Data Gathering in Healthcare Systems.

The existing approach in healthcare entails utilizing data silos found in EHRs, insurance claims, and self-reported patient information. Creating a structure to combine all these data sources is essential to offer a sound foundation for attribution to occur. Utilizing a central database like Microsoft's Fabric for Healthcare for all clinical records will offer extensive data on patients, aiding in the analysis of patient-provider connections. To realize that transformation, the healthcare system needs to embrace contemporary technologies. (Medicare continues to use mainframes…), which have restrictions on resource sharing and their interoperability with other systems. Consequently, transitioning from these systems to sophisticated data collection and sharing platforms such as cloud technologies will offer holistic patient-provider records. Moreover, implementing standards such as Fast Healthcare Interoperability Resources (FHIR) and Health Level Seven (HL7) will facilitate data exchange among healthcare systems. Consequently, there will be a decrease in attribution mistakes and an enhancement in care collaboration.

Embracing Sophisticated Data Analysis

The intricate dynamics of patient-provider relationships, particularly when a patient interacts with several providers, obstruct accurate attribution. Additionally, employing retrospective approaches results in delays in attribution. Sophisticated data analysis, employing machine learning (ML) and artificial intelligence (AI), offers a possible remedy to these challenges. Instruments can assist in analyzing intricate medical data, offering valuable insights. ML models are capable of examining both structured and unstructured data, uncovering patterns that might not be evident with existing analytical models. Traditional attribution models depend on specific methods such as visit counts or location data, whereas advanced algorithms can integrate various datasets into one framework. This includes factors like interaction rates, visit types, diagnostic records, and prescription habits to forecast which provider is probably overseeing the patient's care. This method will diminish misattribution and guarantee that providers accountable for patient outcomes obtain suitable financial rewards. AI can also help diagram intricate patient behaviors, such as frequent changes in providers, to facilitate proactive oversight of their association with particular providers. Moreover, ML algorithms and real-time data analytics enable automated updates for attribution, enhancing accuracy in attribution.

Nevertheless, the success of advanced data analytics and AI in enhancing patient attribution continues to depend on the data quality utilized. Ongoing auditing of data sources is essential when implementing

these technologies to guarantee that precise insights are produced. Effective handling of AI and data analytics technologies can improve accuracy and efficiency in attribution.

Considerations of Ethics

While implementing the aforementioned changes, various ethical considerations must be taken into account. Firstly, the application of AI and ML tools is susceptible to algorithmic bias, leading to discrimination against particular groups or individuals. The conclusions drawn or actions taken by these tools depend on the information they obtain. Consequently, if the data employed in attribution is biased, then the results from ML and AI systems will continue to reflect the existing prejudice. If the problem remains unaddressed, patient attribution will be incorrect, resulting in disparities in healthcare. An effective approach to this issue is performing regular evaluations of automated attribution models to assess their effectiveness across diverse groups to guarantee fairness. Additionally, the requirement for human supervision should not be ignored when utilizing AI and ML tools.

Secondly, patient attribution relies on sensitive health records, which raises ethical issues regarding autonomy and informed consent. Utilizing this information without established guidelines on patient consent violates their privacy rights. Incorporating opt-in or opt-out systems will enable patients to actively consent to or decline the use of their data and to agree with or dismiss designated providers. Patients need to be able to see the kind of information being shared among healthcare systems in order to give their consent. Implementing these practices will ensure a balance between respecting privacy rights and achieving effective attribution in healthcare.

Potential solutions to the challenges in patient attribution involve utilizing advanced data analytics to assess various data sources for precise patient-provider alignment. Additionally, the standardization of data frameworks in healthcare will eliminate the discrepancies in data gathered from various sources. The implementation of standardized guidelines for patient attribution will reduce inconsistencies among paying organizations, alleviating the administrative load for healthcare providers and enabling precise assessment of quality. To meet the shared guidelines, it is essential for all parties to work together in creating attribution models and governance structures.

Nevertheless, prior to implementing any suggested modifications, ethical factors regarding patient privacy and autonomy, along with the potential for bias in automated systems, must be taken into account. Patients ought to have the right to give informed consent regarding the utilization of their medical records and to choose whether to accept or decline a designated provider. All data utilized must be examined to remove any inherent biases, which may exacerbate health disparities. Enhancing patient attribution will result in better value-based care since providers will be responsible for the quality and expenses of care. Patients will gain from enhancements in healthcare, particularly in preventive care. Utilizing a comprehensive solution like Microsoft's Fabric for Healthcare would allow payers like Medicare to concentrate on the principles of Value-Based healthcare rather than the technological infrastructure.

VB-2. How can key themes and technologies such as artificial intelligence, population health analytics, risk stratification, care coordination, usability, quality measurement, and patient engagement be better integrated into APM requirements?

VB-3. What are essential health IT capabilities for value-based care arrangements?

The implementation of varying patient attribution models by distinct payers in the healthcare system was recognized as an obstacle in attribution. Medicare, private insurers, and ACOs employ different attribution rules, causing variations. Certain models determine patient assignments by looking at the provider with the highest number of visits, whereas others depend on patients' self-reports or the primary care provider

indicated on insurance paperwork. These differences can lead to patients being assigned differently based on the payer, leading to administrative difficulties. To tackle this issue, payers must implement standardized attribution criteria, which will aid in evaluating healthcare provider performance within the value-based healthcare framework. Employing a consistent method for attribution will create well-defined criteria for provider eligibility, the selection of prospective or retrospective methodologies, and the incorporation of social determinants of health for adjusting risk in attribution models. Standardization demands strong cooperation between payers and providers to create guidelines that guarantee fairness in attribution and reduce the occurrence of disagreements. The current ambiguity in assessing performance quality will also be resolved because there will be consistency in the anticipated healthcare results among providers.

a. Examples (not comprehensive) may include: care planning, patient event notification, data extraction/normalization, quality performance measurement, access to claims data, attribution and patient ID matching, remote device interoperability, or other patient empowerment tools.

b. What other health IT capabilities have proven valuable to succeeding in value-based care arrangements?

The patient is the key stakeholder in the attribution process as their health is at stake depending on the model applied. Consequently, enhancing patient participation in managing their care will guarantee they obtain their desired quality of treatment. Existing methods, such as utilizing claims data, do not consider the true preferences of patients during later attribution periods. The adoption of self-attribution models will enable patients to take charge of selecting their PCP, and they can make immediate updates if a change becomes necessary. Additionally, the method is consistent with patient-centered care within value-driven healthcare systems. Nonetheless, sufficient patient education is essential for them to grasp the function of a PCP and the significance of attribution.

VB-4. What are the essential data types needed for successful participation in value-based care arrangements?

2. Compliance and Certification

VB-5. In your experience, how do current certification criteria and standards incorporated into the ONC Health IT Certification Program support value-based care delivery?

VB-6. What specific health information technology capabilities that could benefit APMs are not currently addressed by existing certification criteria and standards that should be included under the ONC Health IT Certification Program?

VB-7. How can technology requirements for APMs, established through CEHRT or other pathways, reduce complexity while preserving necessary flexibility?

VB-8. How can other HHS policies supplement CEHRT requirements to better optimize the use of digital health products in APMs? As an example, requirements under the Conditions of Participation for hospitals (42 CFR 482.24(d)) require hospitals to transmit electronic patient event notifications to community providers. What barriers are in place preventing APM participants from receiving the same notifications?

VB-9. What technology requirements should be different for APM organizations when comparing to non-APM organizations (for example, quality reporting, and interoperability)?

VB-10. In the Calendar Year (CY) 2024 Physician Fee Schedule final rule (88 FR 79413), CMS

established that CEHRT requirements for Advanced APMs beyond those in the "Base EHR" definition should be flexible based on what is applicable to the APM that year based on the area of clinical practice. What certification criteria should CMS identify under this flexibility for specific Advanced APMs, or for Advanced APMs in general? Are there specific flexibilities or alternatives to consider for smaller or resource-constrained (such as rural) providers in meeting CEHRT requirements without compromising quality of care or availability of performance data?

3. Technical Standards

VB-11. What specific interoperability challenges have you encountered in implementing value-based care programs?

VB-12. What technology standardization would preserve program-specific flexibility while promoting innovation in APM technology implementation?

VB-13. What improvements to existing criteria and standards would better support value-based care capabilities while reducing provider burden?

VB-14. How could implementing digital identity credentials improve value-based care delivery and outcomes?

VB-15. How could a nationwide provider directory of FHIR endpoints help improve access to patient data and understanding of claims data sources? What key data elements would be necessary in a nationwide FHIR endpoints directory to maximize its effectiveness?

Past Performance

Past Performance 1: CMS Innovation Development and Operation Services (IDOS)	
Customer Name	Centers for Medicare & Medicaid Services (CMS)
Customer/Client POC Email	August R. Nemec III, August.Nemec@cms.hhs.gov
Total Contract Value	$240,158,547
Contract Type	Multi: CPFF and FFP
Period of Performance	1/07/2019 - 6/13/2023
Indication of CPARS available	Yes
Brief Description of Services Provided NewWave served as the prime contractor for the Innovation Development and Operations Services (IDOS) program in support of the Center for Medicare & Medicaid Innovation (CMMI), delivering a full range of services aligned with the RFI's focus on biomedical data processing, analytics, and secure system development. NewWave provided end-to-end software development, data science, and operations support for mission-critical IT systems that process large volumes of structured, semi-structured, and unstructured health-related data. Our team led business analysis and architectural design to support CMMI systems that enable innovation in care models and policy research—similar in complexity and data scale to the translational science and clinical informatics efforts underway at NLM.	

Key areas of relevance include:

- **Big Data Processing & Analytics**: NewWave designed and maintained systems capable of ingesting, classifying, and analyzing high-velocity, high-volume healthcare data. We supported AI/ML-based analytics, integrating data from Electronic Health Records (EHRs), Health Information Exchanges (HIEs), and clinical registries—work highly relevant to LHNCBC's applied research and clinical data science mission.

- **Human-Centered Design & Application Development**: We delivered cloud-based, user-friendly tools (including Salesforce applications) that supported participant management, model solicitation, and reporting. Applications followed usability best practices and were designed to meet diverse user needs across clinical and research communities.

- **Innovation & Emerging Technology**: Through a series of innovation projects, NewWave partnered with CMS to prototype and implement technologies that enable Big Data analytics, improve interoperability, and drive health IT modernization. These projects aligned with emerging policy and business needs, much like NLM's forward-looking approach to scientific computing and biomedical informatics.

- **Quality Analytics & Data Normalization**: Our team led efforts to aggregate, validate, and score quality measures using structured and unstructured healthcare data. We developed tools like the Quality Measures Assessment Tool (QMAT) to support healthcare performance reporting—directly relevant to efforts requiring normalized clinical data analytics at NLM.

- **Cloud-Based Infrastructure & DevOps**: NewWave deployed and managed solutions on secure, scalable cloud platforms that supported high-performance computing and Big Data analytics. Our DevSecOps approach integrated continuous integration, automated testing, and secure release pipelines to ensure system reliability and agility.

- **Security & Compliance**: We ensured full compliance with FISMA, FedRAMP, and CMS-specific data security standards. Our support included production operations, access control, secure data storage, and incident response procedures—mirroring the expectations for contractor staff working with sensitive biomedical data at NIH/NLM.

Through the IDOS program, NewWave demonstrated the capability to deliver secure, cloud-enabled, data-driven IT systems that support research, innovation, and large-scale health data analytics, making it directly applicable as past performance in support of NLM's applied R&D and scientific computing initiatives.